A 2-automatic sequence with non-2-regular abelian complexity

Jean-Paul Allouche, Narad Rampersad, Michel Rigo, Jeffrey Shallit, Manon Stipulanti

arxiv:2610.03837v1; CC0 1.0

Editorial note. One complete original automatic-sequence/nonregular-complexity main theorem. Full mathematical paper90–579, all morphism/factor-count/spectral-parity proofs and original cosine figure retained. Native original biblio.bib/alpha style delegated once to assembly; raw author TeX/archive unchanged. Original CC0 1.0 license preserved exactly.

Editorial note. Original576–577 says part of the paper was obtained through extended discussion with GPT Astra6, original arguments were extensively rewritten, and authors take responsibility. Literal disclosure retained. It does not specify a selected label/core proof discovered by AI; no AI-free label, author certification or dependency-chain investigation is added.

Editorial note. Authors retain responsibility for mathematics. No proof repair, new assumptions, independent proof review, sequence/counterexample search, numerical replay or certification.

Allowed standard prerequisites (editorial).

- Uniform morphisms/codings, automatic and regular sequences, Parikh/abelian complexity, rational cosine classification and irrational rotation density. Original sequence/morphism/coding and regularity definitions retained exactly. Standard automaticity modulo2 and power-subsequence consequences, Niven and Kronecker facts remain original cited prerequisites.. Complete original factor-count and parity/nonregularity argument included; no independent sequence search, numerical graph generation or mathematical certification.

1 Introduction

Combinatorics on words studies finite or infinite sequences taking their values in a given (usually finite) set, called an alphabet. One branch deals with finite sequences where the elements can be unordered, leading to "abelian concepts". Two early occurrences of such notions are [Erd61, p. 240] and [Par66]. For more about this subject and possible applications, see [CC12, FP23].

Here we will address a question about "abelian complexity". Let $\mathbf{x} = (x(n))_{n\geq 0}$ be a sequence over a finite alphabet. The *factor complexity* $\rho_{\mathbf{x}}(n)$ counts the number of distinct length-n blocks appearing in $\mathbf{x}$, while the *abelian complexity* $\rho^{\text{ab}}_{\mathbf{x}}(n)$ counts the number of length-n blocks where the order of the symbols does not matter. Both measures have been studied extensively in the literature; for abelian complexity, see, for example, [RSZ11].

A sequence $\mathbf{x} = (x(n))_{n\geq 0}$ over a finite alphabet is said to be k-automatic if there is a finite deterministic automaton with output (DFAO) that, on input n expressed in base $k \geq 2$, terminates in a state with output $x(n)$. Equivalently ([Cob72] and [AS03, Thm. 6.3.2]), a sequence is k-automatic if and only if it is the image, under a coding, of a fixed point of a k-uniform morphism (i.e., the image of each letter has length equal to k). The automatic sequences form a large and interesting class that has been studied for more than 50 years [Cob72, AS03].

The k-regular sequences are a generalization of k-automatic sequences [AS92, AS03]. We say that a sequence $\mathbf{x} = (x(n))_{n\geq 0}$ over $\mathbb{Z}$ is k-regular if its k-kernel

$$\mathcal{K}_k(\mathbf{x}) := \{(x(k^i n + e))_{n\geq 0} \,:\, i \geq 0,\ 0 \leq e < k^i\}$$

spans a finite-dimensional vector space over $\mathbb{Q}$.

It is known that the factor complexity of a k-automatic sequence is k-regular [CD10, Prop. 4.3] (a weaker result had been proven earlier by Mossé [Mos96]). A question in Madill and Rampersad [MR13] (to which the answer was widely conjectured to be yes) asked whether the same is true for the abelian complexity of a k-automatic sequence. This holds for several sequences that have been investigated, for example, successively:

- the Thue–Morse sequence, whose abelian complexity is periodic, and hence k-regular for every k [RSZ11, Thm. 3.1]. That early overview also contains two nice results, namely [RSZ11, Thm. 1.7 and Thm. 3.3]. We leave to the reader the pleasure of applying those two theorems to construct k-automatic sequences whose abelian complexity is periodic, and hence k-regular for every k.
- the regular paperfolding sequence [MR13].
- fixed points of k-uniform morphisms φ on $\{0, 1\}$ such that $\big||\varphi(0)|_0 - |\varphi(1)|_0\big| \neq 1$, which is a consequence of [BSSW16, Cor. 1].
- the Golay–Shapiro sequence (aka the Rudin–Shapiro sequence) [LCWW17].
- the Thue–Morse sequence over a ternary alphabet, which is an easy consequence of [KK17, Thm. 4.1].
- the Cantor sequence [CLW18, Thm. 2].
- the generalized Thue–Morse sequences, which is an easy consequence of [CW19, Cor. 1].
- automatic sequences satisfying certain extra conditions [CW19, Thm. 2].

- automatic sequences $\mathbf{s}$ such that the Parikh vectors of the length-n prefixes of $\mathbf{s}$ form a synchronized sequence and such that the abelian complexity of $\mathbf{s}$ is bounded above by a constant [Sha21].
- automatic sequences that are fixed points of Parikh-collinear morphisms [RSW24, Thm. 3].
- the 2-block coding of generalized Cantor sequences $\mathbf{c}$ that have mirror-invariant sets of factors [LCW22, Thm. 1].

An even stronger conjecture appears in [PRRV15]: namely, that *the ℓ-abelian complexity of a k-automatic sequence is k-regular.* Several papers give positive answers to this question; see [CLW18, CDO+, Gre15, LCW22, PRRV15].

However, in this note we give an example of a k-automatic sequence whose abelian complexity is not k-regular (see Theorem 7). This result also refutes a conjecture of Popoli and two of the present authors [PSS24, Conj. 4] about additive complexity, since for binary sequences additive complexity and abelian complexity coincide.

2 Construction and main theorem

Let $\mathbb{N} = \{0, 1, 2, \ldots\}$. A *factor* of an infinite word is a finite contiguous subword appearing in it. For a finite word u, let $|u|$ be its length and $|u|_a$ the number of occurrences of the letter a. The *abelian complexity* $\rho^{\mathrm{ab}}_{\mathbf{x}}(n)$ of a binary word $\mathbf{x}$ then counts the distinct pairs $(|u|_0, |u|_1)$ among its length-n factors [RSZ11].

Lemma 1. *Let $\mathbf{x}$ be an infinite binary word. For all $n \geq 0$, we let M_n and L_n, respectively, denote the maximum and minimum numbers of 1's in length-n factors of $\mathbf{x}$. Then, for all $n \geq 0$, we have $\rho^{\mathrm{ab}}_{\mathbf{x}}(n) = M_n - L_n + 1$.*

Proof. Sliding a window one position changes its count by at most 1, so all intermediate counts occur. □

Remark 2. For binary sequences the quantity $M_n - L_n$ is exactly the balance function of Adamczewski [Ada03].

For $n \in \mathbb{N}$, define $d_3(n)$ to be the number of occurrences of the digit 3 in the base-4 expansion of n, i.e.,

$$\text{if } n = \sum_{i=0}^{\ell} n_i\, 4^i \text{ with } n_0, \ldots, n_\ell \in \{0, 1, 2, 3\}, \text{ then } d_3(n) = |n_\ell \cdots n_0|_3.$$

Define the sequences $\mathbf{s} = (s(n))_{n\geq 0}$ and $\mathbf{b} = (b(n))_{n\geq 0}$ by

$$s(n) = d_3(n) \bmod 3 \quad \text{and} \quad b(n) = \begin{cases} 1, & \text{if } s(n) = 0; \\ 0, & \text{if } s(n) \neq 0. \end{cases}$$

The first few values of these sequences can be found in Table 1.

$\mathbf{x}$	$x(0)x(1)x(2)\cdots x(63)$
$\mathbf{d}_3$	0001000100011112000100010001111200010001000111121112111211122223
$\mathbf{s}$	0001000100011112000100010001111200010001000111121112111211122220
$\mathbf{b}$	1110111011100000111011101110000011101110111000000000000000000001

Table 1: The values of the sequences $\mathbf{d}_3$, $\mathbf{s}$, and $\mathbf{b}$ for $0 \le n \le 63$.

Note that $(d_3(n))_{n\ge 0}$ and $(s(n))_{n\ge 0}$ differ for the first time at $n = 63 = 4^3 - 1$, for which we have $d_3(63) = 3$ and thus $s(63) = 0$. The binary word $\mathbf{b}$ records the positions of 0 in $\mathbf{s}$ (also known as the characteristic sequence of 0's in $\mathbf{s}$).

Since $d_3(4n + j) = d_3(n)$ for $j < 3$ and $d_3(4n + 3) = d_3(n) + 1$ for all $n \ge 0$, we have the following result.

Lemma 3. *The sequence* $\mathbf{s}$ *is the fixed point starting with* 0 *of the morphism* $\sigma\colon \{0,1,2\}^* \to \{0,1,2\}^*, a \mapsto aaa(a+1)$*, where letters are added modulo* 3*.*

Remark 4. It may be interesting to study the morphisms σ_r, $r \ge 1$, where $a \mapsto a^r(a+1)$ and letters are taken mod r. When $r = 1$ we obtain the Thue–Morse morphism for which the corresponding sequence is 2-automatic and its abelian complexity is 2-regular, as mentioned previously. When $r = 3$ and $A = \{0,1,2\}$, we obtain the morphism of Lemma 3 for which we have Theorem 7. What can be said for other values of r and A?

With σ one can associate the matrix $\mathcal{M}_\sigma$ in $\mathbb{N}^{3\times 3}$ where $(\mathcal{M}_\sigma)_{i,j} = |\sigma(j)|_i$ for $i, j \in \{0,1,2\}$, i.e.,

$$\mathcal{M}_\sigma = \begin{pmatrix} 3 & 0 & 1 \\ 1 & 3 & 0 \\ 0 & 1 & 3 \end{pmatrix}. \tag{2.1}$$

Its eigenvalues are 4, λ and $\overline{\lambda}$, where $\lambda = \frac{1}{2}\left(5 + i\sqrt{3}\right)$.

Lemma 5. *Every pair of letters in* $\{0,1,2\}$ *occurs in* $\mathbf{s}$*.*

Proof. Observe that all three letters occur in $\sigma^2(0) = 0001000100011112$. Now, for $a \in \{0,1,2\}$, the pairs aa and $a(a+1)$ occur inside $\sigma(a)$, and $(a+1)a$ occurs across the boundary in $\sigma(aa)$. □

Lemma 6. *The sequence* $\mathbf{b}$ *is* 2*-automatic.*

Proof. The sequence $\mathbf{s}$ is 4-automatic by Lemma 3, as the morphism σ is 4-uniform. Since $\mathbf{b}$ is obtained from $\mathbf{s}$ by applying the coding $0 \mapsto 1; 1, 2 \mapsto 0$, the sequence $\mathbf{b}$ is also 4-automatic. It is known that, for $m \ge 1$, a sequence is k-automatic if and only if it is k^m-automatic; see, for example, [AS03, Thm. 6.6.4]. In particular, the sequence $\mathbf{b}$ is 2-automatic. □

We now formulate our main contribution.

Theorem 7. *The sequence* $\mathbf{b}$ *is* 2*-automatic, while its abelian complexity* $\rho_{\mathbf{b}}^{\mathrm{ab}}$ *is not* 2*-regular.*

3 An exact formula for a subsequence of $\rho_{\mathbf{b}}^{\mathrm{ab}}$

For all $m \in \mathbb{N}$, write $N_m = (N_m(0), N_m(1), N_m(2))$, where $N_m(a) = |\sigma^m(0)|_a$, i.e., $N_m^\top$ is the first column of the matrix $(\mathcal{M}_\sigma)^m$. Observe that $\sum_{a=0}^{2} N_m(a) = |\sigma^m(0)| = 4^m$. Put

$$g(m) = \max_a N_m(a) - \min_a N_m(a). \tag{3.1}$$

These counts satisfy

$$N_0 = (1,0,0), \qquad N_{m+1}(a) = 3N_m(a) + N_m(a-1). \tag{3.2}$$

The first few values of the vectors $(N_m)_{m\geq 0}$ are

$$(1,0,0), (3,1,0), (9,6,1), (28,27,9), (93,109,54), (333,420,271),$$

so the first few values of $(g(m))_{m\geq 0}$ are $1, 3, 8, 19, 55, 149$. We will prove the following exact formula for the sequence $(\rho_{\mathbf{b}}^{\mathrm{ab}}(4^m))_{m\geq 0}$, whose first few values are

$$2,\ 5,\ 14,\ 38,\ 95,\ 260,\ 707,\ 1787,\ 4805,\ 13112.$$

Proposition 8. *For all $m \geq 0$, set $a(m) = \rho_{\mathbf{b}}^{\mathrm{ab}}(4^m)$. Then, for all $m \geq 0$, we have*

$$a(m+1) - a(m) = 3g(m) \quad \text{and} \quad a(m) = 2 + 3\sum_{j=0}^{m-1} g(j),$$

where the sequence $(g(m))_{m\geq 0}$ is defined in (3.1). In particular, $a(m) \equiv 2 \pmod 3$ for all $m \geq 0$.

To prove this result, we need a lemma.

Lemma 9. *To each letter $a \in \{0,1,2\}$, assign a real weight $\mathrm{wt}(a)$. Define the weight of a word $z = z_1 \cdots z_\ell$ to be the sum of its letter weights, i.e., $\mathrm{wt}(z) = \sum_{i=1}^{\ell} \mathrm{wt}(z_i)$. For $a \in \{0,1,2\}$ and $m \geq 0$, let $\mathrm{wt}_m(a)$ be the weight of $\sigma^m(a)$, and let H_m be the greatest weight of a length-4^m factor of $\mathbf{s}$. Then, for all $m \geq 0$,*

$$H_{m+1} = H_m + 3\max_a \mathrm{wt}_m(a).$$

Proof. Take $m \geq 0$. By Lemma 3, $\mathbf{s}$ is a fixed point of a 4-uniform morphism, so every length-4^{m+1} factor consists of a suffix of one length-4^m substitution block, three complete blocks, and a prefix of a fifth block. The suffix and prefix together have length 4^m and form a factor of $\mathbf{s}$, since every pair of letters occurs by Lemma 5. Their weight is at most H_m, which shows that $H_{m+1} \leq H_m + 3\max_a \mathrm{wt}_m(a)$.

For equality, for each $m \geq 0$, choose a letter c_m maximizing wt_m, i.e.,

$$\mathrm{wt}_m(c_m) = \max_a \mathrm{wt}_m(a).$$

Since $\sigma^{m+1}(a) = \sigma^m(aaa(a+1))$, we have $\mathrm{wt}_{m+1}(a) = 3\mathrm{wt}_m(a) + \mathrm{wt}_m(a+1)$. Recall that letters are taken modulo 3; the alphabet is $\mathbb{Z}/3\mathbb{Z}$. We can choose the letters of the sequence $(c_m)_{m\geq 0}$ successively with $c_{m+1} \in \{c_m - 1, c_m\}$. Indeed, we have

$$\mathrm{wt}_{m+1}(c_m) - \mathrm{wt}_{m+1}(c_m+1) = 3\mathrm{wt}_m(c_m) - 2\mathrm{wt}_m(c_m+1) - \mathrm{wt}_m(c_m-1) \geq 0,$$

where we used that $c_m + 2 = c_m - 1$. The inequality $\mathrm{wt}_{m+1}(c_m) \geq \mathrm{wt}_{m+1}(c_m+1)$ implies that c_{m+1} maximizing wt_{m+1} can be chosen in $\{c_m - 1, c_m\}$.

We prove inductively that H_m is attained by a factor starting in the first block of either $\sigma^m(c_m-1)\sigma^m(c_m)$ or $\sigma^m(c_m)\sigma^m(c_m+1)$. In other words, there exist words p, s, x, y such that sp is a length-4^m factor of maximal weight, with s nonempty, $\sigma^m(c_m-1) = xs$ and $\sigma^m(c_m) = py$; or $\sigma^m(c_m) = xs$ and $\sigma^m(c_m+1) = py$.

For the base case $m = 0$, take the single letter c_0, at the start of $c_0(c_0+1)$, which occurs in $\mathbf{s}$ by Lemma 5. This has maximum weight by definition.

Now turn to the induction step. Write $c = c_m$, $U_a = \sigma^m(a)$. The word

$$\sigma^{m+1}(c-1)\,\sigma^{m+1}(c) = U_{c-1}\,U_{c-1}\,U_{c-1}\,U_c\,U_c\,U_c\,U_c\,U_{c+1} \tag{3.3}$$

is a factor of $\mathbf{s}$ since $(c-1)c$ occurs in $\mathbf{s}$ by Lemma 5. By the induction hypothesis, there is a maximizing factor z of length 4^m starting in the first block of either $U_{c-1}U_c$ or U_cU_{c+1}. We consider two cases. For a word w, let $\mathrm{pref}_t(w)$ (resp., $\mathrm{suff}_t(w)$) be the length-t prefix (resp., suffix) of w.

Case 1. Suppose that z starts in the first block of $U_{c-1}U_c$. We have $z = \mathrm{suff}_{4^m-t}(U_{c-1})\mathrm{pref}_t(U_c)$ for some $0 \leq t < 4^m$. In the length-$2\cdot 4^{m+1}$ factor highlighted in (3.3), consider the factor $\mathrm{suff}_{4^m-t}(U_{c-1})\,U_c\,U_c\,U_c\,\mathrm{pref}_t(U_c)$ starting in position $2\cdot 4^m + t$. Since the weight function is additive, the weight of this length-4^{m+1} factor is

$$\begin{aligned}\mathrm{wt}(\mathrm{suff}_{4^m-t}(U_{c-1})) + \mathrm{wt}(U_cU_cU_c) + \mathrm{wt}(\mathrm{pref}_t(U_c)) &= \mathrm{wt}(z) + 3\mathrm{wt}(U_c)\\ &= H_m + 3\max_a \mathrm{wt}_m(a),\end{aligned}$$

by definition of c and z. Since we have found a length-4^{m+1} factor with this weight, we conclude that $H_{m+1} \geq H_m + 3\max_a \mathrm{wt}_m(a)$.

Case 2. Suppose that z starts in the first block of U_cU_{c+1}. We have $z = \mathrm{suff}_{4^m-t}(U_c)\,\mathrm{pref}_t(U_{c+1})$ for some $0 \leq t < 4^m$. In the length-$2\cdot 4^{m+1}$ factor highlighted in (3.3), consider the factor $\mathrm{suff}_{4^m-t}(U_c)\,U_c\,U_c\,U_c\,\mathrm{pref}_t(U_{c+1})$ starting in position $3\cdot 4^m + t$. Similarly to the first case, its weight is $H_m + 3\max_a \mathrm{wt}_m(a)$. □

We turn to the proof of Proposition 8.

Proof of Proposition 8. For all $n \geq 0$, let M_n and L_n, respectively, denote the maximum and minimum numbers of 1's in length-n factors of $\mathbf{b}$. By Lemma 1, we have

$$a(m) = \rho^{\mathrm{ab}}_{\mathbf{b}}(4^m) = M_{4^m} - L_{4^m} + 1 \tag{3.4}$$

for all $m \geq 0$. To obtain the first formula of the statement, we apply Lemma 9 to two different sequences of weights.

First, apply Lemma 9 to the weights $\mathrm{wt}(0) = 1$ and $\mathrm{wt}(1) = \mathrm{wt}(2) = 0$. This particular weight simply counts the number of 0's in words. With the notation of this lemma, H_m then gives the maximum number of 0's in length-4^m factors of $\mathbf{s}$, which is M_{4^m} since $\mathbf{b}$ is the characteristic sequence of 0's in $\mathbf{s}$. Now we examine the weight $\mathrm{wt}_m(a)$ of $\sigma^m(a)$. Note that each letter $0, 1, 2$ in $\sigma^m(0)$ corresponds to the letter $a, a+1, a+2$ (if we count modulo 3) in $\sigma^m(a)$. Therefore, the counts of $0, 1, 2$ in $\sigma^m(a)$ are permutations of the counts of $0, 1, 2$ in $\sigma^m(0)$. In other words, the columns of $(\mathcal{M}_\sigma)^m$ are cyclic permutations of each other. Consequently, recalling the definition of $N_m(a) = |\sigma^m(0)|_a$ at the beginning of the section, Lemma 9 with $H_{m+1} = H_m + 3\max_a \mathrm{wt}_m(a)$ yields the recurrence relation

$$M_{4^{m+1}} = M_{4^m} + 3\max_a N_m(a).$$

Second, applying Lemma 9 to the weights $\mathrm{wt}(0) = -1$ and $\mathrm{wt}(1) = \mathrm{wt}(2) = 0$ gives in a similar way

$$L_{4^{m+1}} = L_{4^m} + 3\min_a N_m(a).$$

Together with (3.4), these recurrence relations give

$$\begin{aligned} a(m+1) = M_{4^{m+1}} - L_{4^{m+1}} + 1 &= M_{4^m} - L_{4^m} + 1 + 3\left(\max_a N_m(a) - \min_a N_m(a)\right) \\ &= a(m) + 3g(m), \end{aligned}$$

which is the first equality of the statement. For the second, use this recurrence repeatedly and recall that $a(0) = 2$. □

4 Parity and non-regularity

In this section, we consider the complex numbers

$$\omega = e^{2\pi i/3} = \frac{-1}{2} + i\frac{\sqrt{3}}{2} \quad \text{and} \quad \lambda = 3 + \omega = (5 + i\sqrt{3})/2 = \sqrt{7}e^{i\theta},$$

where $\theta = \arctan(\sqrt{3}/5)$ is the argument of λ (recall that λ is an eigenvalue of $\mathcal{M}_\sigma$ defined in (2.1)).

Lemma 10. *We have* $\sum_{a=0}^{2} N_m(a) = 4^m$ *and* $\sum_{a=0}^{2} \omega^a N_m(a) = \lambda^m$.

Proof. The first equality follows by definition, as we have

$$\sum_{a=0}^{2} N_m(a) = \sum_{a=0}^{2} |\sigma^m(0)|_a = |\sigma^m(0)| = 4^m.$$

For the second equality, we proceed by induction on $m \geq 0$. The case $m = 0$ is clear, since $N_0 = (1, 0, 0)$ and for $m = 1$, we have $N_1 = (3, 1, 0)$, so

$$\sum_{a=0}^{2} \omega^a N_1(a) = \omega^0 \cdot 3 + \omega \cdot 1 + \omega^2 \cdot 0 = \lambda$$

by definition. Now assume that the second equality holds for $m \geq 0$. Using the recurrence given in (3.2), the fact that $\omega^{-1} = \omega^2$ and the induction hypothesis, we get

$$\sum_{a=0}^{2} \omega^a N_{m+1}(a) = 3 \sum_{a=0}^{2} \omega^a N_m(a) + \omega \sum_{a=0}^{2} \omega^{a-1} N_m(a-1) = (3+\omega) \cdot \lambda^m = \lambda^{m+1},$$

as desired. □

Since the matrix $\mathcal{M}_\sigma$ has only simple eigenvalues, it is diagonalizable. It is therefore straightforward to express the entries of $(\mathcal{M}_\sigma)^m$ as linear combinations of the mth powers of these eigenvalues. In particular, a few elementary manipulations yield the vector N_m.

Lemma 11. *For each letter $a \in \{0,1,2\}$, we have*

$$N_m(a) = \frac{4^m + 2 \cdot 7^{m/2} \cos\left(m\theta - \frac{2\pi}{3}a\right)}{3}.$$

Proof. Using the first equality in Lemma 10 and the two equalities derived from identifying the real and imaginary parts in the second equality of Lemma 10, we obtain a system of three equations in the unknowns $N_m(0), N_m(1), N_m(2)$, whose solution is

$$N_m(0) = \frac{1}{3}\left(4^m + 2 \cdot 7^{m/2} \cos(m\theta)\right), \tag{4.1}$$

$$N_m(1) = \frac{1}{3}\left(4^m - 7^{m/2} \cos(m\theta) + \sqrt{3} \cdot 7^{m/2} \sin(m\theta)\right), \tag{4.2}$$

$$N_m(2) = \frac{1}{3}\left(4^m - 7^{m/2} \cos(m\theta) - \sqrt{3} \cdot 7^{m/2} \sin(m\theta)\right). \tag{4.3}$$

Recalling the formula $\cos(x-y) = \cos(x)\cos(y) + \sin(x)\sin(y)$ and the particular values $\cos(\frac{2\pi}{3}) = \frac{-1}{2}$, $\sin(\frac{2\pi}{3}) = \frac{\sqrt{3}}{2}$, this gives the formula in the statement. □

Lemma 12. *The real number $\frac{\theta}{\pi}$ is irrational.*

Proof. Using the values $N_2(0) = 9$ (see the first few values of $(N_m)_{m\geq 0}$ given after (3.2)) and the equality in Lemma 11 for $a = 0$ and $m = 2$, we get $\cos(2\theta) = \frac{11}{14} \in \mathbb{Q}$. Niven's theorem (see [PV21, Thm. 1, p. 132] and the discussion there) restricts rational cosines of rational multiples of π to $\{0, \pm\frac{1}{2}, \pm 1\}$. This implies that 2θ is not a rational multiple of π, which is enough. □

The next lemma determines the parity of the sequence $g(3n)$ in terms of $|\cos(3n\theta)|$. The first few values of the latter sequence are shown in Figure 1.

Lemma 13. *For all $n \geq 1$, we have $g(3n) \equiv 0 \pmod 2$ if and only if $|\cos(3n\theta)| < 1/2$.*

Proof. Taking the recurrence (3.2) modulo 2 shows that the sequence $(N_m)_{m\geq 1}$ cycles through the three vectors $(1,1,0), (1,0,1), (0,1,1)$ (note that $N_0 = (1,0,0)$). Thus, for $n \geq 1$, $N_{3n}(0)$ is even and the other two counts $N_{3n}(1), N_{3n}(2)$ are odd. From (3.1), their range $g(3n)$ is even exactly when $N_{3n}(0)$ lies strictly between the other two quantities $N_{3n}(1), N_{3n}(2)$. Indeed,

the only orderings of the natural numbers $N_{3n}(0), N_{3n}(1), N_{3n}(2)$ giving an even range are when $N_{3n}(1) < N_{3n}(0) < N_{3n}(2)$ or $N_{3n}(2) < N_{3n}(0) < N_{3n}(1)$. Lemma 11 gives

$$\begin{aligned}(N_m(0) - N_m(1))\,(N_m(0) - N_m(2)) &= \frac{7^m}{9}\Big(3\cos(m\theta) - \sqrt{3}\sin(m\theta)\Big)\Big(3\cos(m\theta) + \sqrt{3}\sin(m\theta)\Big)\\ &= \frac{7^m}{3}\big(3\cos^2(m\theta) - \sin^2(m\theta)\big)\\ &= \frac{7^m}{3}\big(4\cos^2(m\theta) - 1\big).\end{aligned}$$

Consequently,

$$g(3n) \equiv 0 \pmod 2 \quad\Longleftrightarrow\quad \frac{7^{3n}}{3}\big(4\cos^2(3n\theta) - 1\big) < 0 \quad\Longleftrightarrow\quad |\cos(3n\theta)| < 1/2, \quad (4.4)$$

as desired. □

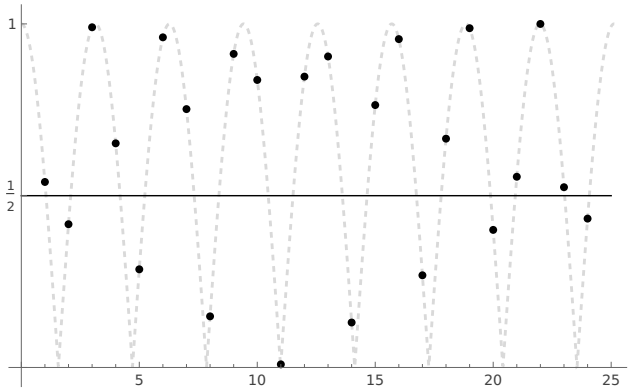


Figure 1: The first few values of $(|\cos(3n\theta)|)_{n\geq 1}$.

Recall that a sequence $(x(n))_{n\geq 0}$ is *eventually periodic* if $x(n+p) = x(n)$ for some $p \geq 1$ and all sufficiently large n.

Proposition 14. *The sequence* $(\rho_{\mathbf{b}}^{\mathrm{ab}}(4^m) \bmod 2)_{m\geq 0}$ *is not eventually periodic.*

Proof. For the sake of conciseness, set $a(m) = \rho_{\mathbf{b}}^{\mathrm{ab}}(4^m)$ for all $m \geq 0$.

Recall that Lemma 12 implies that θ is not a rational multiple of π. Fix a positive integer p. The angles $(3pn\theta)_{n\geq 1}$ are dense modulo π, even after deleting finitely many terms. This is the density theorem for irrational rotations (Kronecker's theorem); see, e.g., [HW79, Thm. 439, p. 376]. The conditions $|\cos t| < 1/2$ and $|\cos t| > 1/2$ each hold on nonempty open intervals. Lemma 13 implies that $(g(3pn) \bmod 2)_{n\geq 1}$ takes both values infinitely often. It follows that $(g(m) \bmod 2)_{m\geq 0}$ cannot be eventually periodic with any period p. By Proposition 8, the same is true of $(a(m) \bmod 2)_{m\geq 0}$, since eventual periodicity is preserved by taking first differences. This proves the statement. □

We have everything in our hands to prove the main theorem.

Proof of Theorem 7. The fact that the sequence $\mathbf{b}$ is 2-automatic follows from Lemma 6. Now assume, to the contrary, that $\rho_{\mathbf{b}}^{\mathrm{ab}}$ is 2-regular. By [AS03, Cor. 16.1.6], the sequence $(\rho_{\mathbf{b}}^{\mathrm{ab}}(n) \bmod 2)_{n\geq 0}$ is 2-automatic.

Observe that, if a sequence $(x(n))_{n\geq 0}$ is 2-automatic, then the sequence $(x(4^n))_{n\geq 0}$ is eventually periodic. Indeed, there is a DFAO $\mathcal{A}$ that, when fed with the binary expansion of n, outputs $x(n)$. Since the binary expansion of 4^n is $1(00)^n$ for every $n \geq 0$, the sequence of states $(q_0 \cdot 1(00)^n)_{n\geq 0}$ reached from the initial state of $\mathcal{A}$ is eventually periodic. As the state set is finite, this sequence is eventually periodic, and hence so is the corresponding output sequence $(x(4^n))_{n\geq 0}$.

Applying this to our case, the sequence $(\rho_{\mathbf{b}}^{\mathrm{ab}}(4^n) \bmod 2)_{n\geq 0}$ is eventually periodic, contradicting Proposition 14. □

Acknowledgments

Narad Rampersad is supported by NSERC grant RGPIN-2025-04076. Michel Rigo is supported by the FNRS grant T.196.23 (PDR). Jeffrey Shallit is supported by NSERC grant RGPIN-2024-03725. Manon Stipulanti is an FNRS Research Associate (CQ) supported by the Research grant 1.C.104.24F.

Declaration of AI Usage

Part of this paper was obtained through extended discussion with GPT Astra 6, a large language AI model. We have extensively rewritten the original arguments. The authors take responsibility for all of the contents.

References

[Ada03] Boris Adamczewski. Balances for fixed points of primitive substitutions. *Theoret. Comput. Sci.*, 307:47–75, 2003.

[AS92] Jean-Paul Allouche and Jeffrey Shallit. The ring of k-regular sequences. *Theoret. Comput. Sci.*, 98(2):163–197, 1992.

[AS03] Jean-Paul Allouche and Jeffrey Shallit. *Automatic Sequences: Theory, Applications, Generalizations*. Cambridge University Press, Cambridge, 2003.

[BSSW16] Francine Blanchet-Sadri, Daniel Seita, and David Wise. Computing abelian complexity of binary uniform morphic words. *Theoret. Comput. Sci.*, 640:41–51, 2016.

[CC12] Manolis Christodoulakis and Michalis Christou. Abelian concepts in strings: a review. In Jan Holub, Bruce W. Watson, and Jan Žďárek, editors, *Festschrift for Bořivoj Melichar*, pages 19–45. Czech Technical University, Prague, 2012. Available at https://www.stringology.org/papers/Festschrift_BM70.pdf.

[CD10] Arturo Carpi and Valerio D'Alonzo. On factors of synchronized sequences. *Theoret. Comput. Sci.*, 411(44-46):3932–3937, 2010.

[CDO+] Jean-Michel Couvreur, Martin Delacourt, Nicolas Ollinger, Pierre Popoli, Jeffrey Shallit, and Manon Stipulanti. Effective computation of generalized abelian complexity for Pisot type substitutive sequences. Arxiv preprint arXiv:2504.13584 [cs.FL], April 22 2025. Available at https://arxiv.org/abs/2504.13584.

[CLW18] Jin Chen, Xiao-Tao Lü, and Wen Wu. On the k-abelian complexity of the Cantor sequence. *J. Combin. Theory Ser. A*, 155:287–303, 2018.

[Cob72] Alan Cobham. Uniform tag sequences. *Math. Systems Theory*, 6:164–192, 1972.

[CW19] Jin Chen and Zhi-Xiong Wen. On the abelian complexity of generalized Thue-Morse sequences. *Theoret. Comput. Sci.*, 780:66–73, 2019.

[Erd61] Paul Erdős. Some unsolved problems. *Magyar Tud. Akad. Mat. Kutató Int. Közl.*, 6:221–254, 1961.

[FP23] Gabriele Fici and Svetlana Puzynina. Abelian combinatorics on words: A survey. *Computer Sci. Rev.*, 47:100532, 2023.

[Gre15] Florian Greinecker. On the 2-abelian complexity of the Thue-Morse word. *Theoret. Comput. Sci.*, 593:88–105, 2015.

[HW79] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*. Oxford University Press, 5th edition, 1979.

[KK17] Idrissa Kaboré and Boucaré Kientéga. Abelian complexity of Thue-Morse word over a ternary alphabet. In *Combinatorics on words*, volume 10432 of *Lecture Notes in Comput. Sci.*, pages 132–143. Springer, Cham, 2017.

[LCW22] Xiao-Tao Lü, Jin Chen, and Zhi-Xiong Wen. On the 2-abelian complexity of generalized Cantor sequences. *Theoret. Comput. Sci.*, 936:172–183, 2022.

[LCWW17] Xiao-Tao Lü, Jin Chen, Zhi-Xiong Wen, and Wen Wu. On the abelian complexity of the Rudin-Shapiro sequence. *J. Math. Anal. Appl.*, 451(2):822–838, 2017.

[Mos96] Brigitte Mossé. Reconnaissabilité des substitutions et complexité des suites automatiques. *Bull. Soc. Math. France*, 124(2):329–346, 1996.

[MR13] Blake Madill and Narad Rampersad. The abelian complexity of the paperfolding word. *Discrete Math.*, 313(7):831–838, 2013.

[Par66] Rohit J. Parikh. On context-free languages. *J. Assoc. Comput. Mach.*, 13:570–581, 1966.

[PRRV15] Aline Parreau, Michel Rigo, Eric Rowland, and Élise Vandomme. A new approach to the 2-regularity of the ℓ-abelian complexity of 2-automatic sequences. *Electron. J. Combin.*, 22(1):Paper 1.27, 44, 2015.

[PSS24] Pierre Popoli, Jeffrey Shallit, and Manon Stipulanti. Additive word complexity and Walnut. In Siddharth Barman and Sławomir Lasota, editors, *44th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2024)*, volume 323 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 32:1–32:18, Dagstuhl, Germany, 2024. Schloss Dagstuhl–Leibniz-Zentrum für Informatik.

[PV21] Bonaventura Paolillo and Giovanni Vincenzi. On the rational values of trigonometric functions of angles that are rational in degrees. *Math. Mag.*, 94(2):132–134, 2021.

[RSW24] Michel Rigo, Manon Stipulanti, and Markus A. Whiteland. Automatic abelian complexities of Parikh-collinear fixed points. *Theory Comput. Syst.*, 68(6):1622–1639, 2024.

[RSZ11] Gwénaël Richomme, Kalle Saari, and Luca Q. Zamboni. Abelian complexity of minimal subshifts. *J. Lond. Math. Soc. (2)*, 83(1):79–95, 2011.

[Sha21] Jeffrey Shallit. Abelian complexity and synchronization. *INTEGERS*, 21:Paper No. A36, 2021.